

Technology Stack

Date	02 July 2026
Team ID	xxxxxx
Project Name	Emotion Detection & Learning Support Engine
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	Streamlit, HTML, CSS	Streamlit was chosen to rapidly develop an interactive and responsive web interface with minimal code. Custom HTML & CSS were used to enhance UI appearance and improve user experience.

2	Backend / Server-Side	Python	Python is chosen for backend development because of its simplicity, vast libraries, fast development, and massive community support.
3	Database / Data Storage	JSON Files(bert_label_encoder.json, special_tokensL_map.json, tokenizer.json, tokenizer_config.json)	JSON files were used for lightweight local storage of conversation history and user feedback without requiring a dedicated database server.
4	Cloud / Hosting / Deployment	Local Deployment using Streamlit	The application is currently deployed locally for development and demonstration purposes. The architecture can be easily migrated to cloud platforms in future versions.
5	Version Control & CI/CD	Git, GitHub	Git & GitHub were used for source code management, version tracking, collaboration, and project backup
6	Third-Party APIs / Other Tools	Gemini API, Hugging Face Transformers	Gemini API enables fact verification, Hugging Face Transformers provide NLP capabilities for event analysis.